DTP Market Value Confidence Engine (MVCE) Full Product Documentation

1. Product Overview

Product Name: DTP Market Value Confidence Engine (MVCE)

Purpose: AI-assisted football valuation prediction module designed to estimate likely first Transfermarkt-style market value bands for Nigerian grassroots football players.

2. Core Philosophy

The engine should emulate real-world market logic where visibility, verification, competitive level, development upside, exposure, and demand influence valuation — not just raw statistics.

3. Product Objectives

Predict likely first market value bands Help scouts identify undervalued talent Standardize player visibility assessment Increase trust for international scouts

4. User Types

Scout, Academy Admin, DTP Admin, Analyst, and Club Recruiter.

5. System Architecture

Frontend Modules: Player Input Form Player Comparison Engine Valuation Dashboard Confidence Analyzer Scout Insights Panel **Backend Services:** Scoring Engine Confidence Engine Rule Engine Comparable Player Engine

6. Core Product Flow

Player Created → Variables Entered → Scoring Engine Runs → Multipliers Applied → Confidence Band Calculated → Market Value Range Generated

7. Core Input Variables

Player Profile: age, position, nationality, dominant foot, height, weight

Competition: club type, competition level, academy reputation

Performance: goals, assists, tackles, dribbles, interceptions

DTP Metrics: sprint speed, accelerations, turning speed, distance with ball, top speed, explosive metrics

Visibility Signals: verified footage, scout reports, tournament exposure, social visibility

Market Signals: club interest, youth national team, transfer rumors, professional contract

8. Scoring Engine

The engine uses a weighted 100-point scoring system. Age Upside – 15% Competition Level – 15% Match Experience – 15% Performance Quality – 12% Visibility Signals – 10% Market Signals – 12% Contract Security – 6% DTP Athletic Metrics – 10% Risk Adjustment – -15%

9. Confidence Bands

ScoreConfidence 0–24Very Low 25–39Low 40–54Moderate 55–69Strong 70–84Very Strong 85–100Elite Prospect

10. Market Value Bands

Score RangeMarket Value 0–24No Reliable Value 25–39€10k–€25k 40–54€25k–€75k 55–69€75k–€150k 70–84€150k–€350k 85–100€350k–€750k+

11. Multipliers

Verification Multiplier:
No proof: 0.6
Video only: 0.8
Video + match sheets: 1.0
Video + stats + verified reports: 1.15
National team proof: 1.25

Market Demand Multiplier:
Local club: 1.05
NPFL: 1.15
African club: 1.3
European club: 1.6
Elite European club: 2.0

12. Position Multipliers

GK: 0.85
CB: 1.0
FB/WB: 1.05
DM: 1.1

CM: 1.15
AM: 1.2
Winger: 1.35
Striker: 1.4

13. Comparable Player Engine

The engine compares players based on: Age Position Height Competition level Athletic profile Visibility profile Market signals

14. Dashboard UI

Main Layout:
Left Panel: Player profile and bio
Center Panel: Confidence score and predicted value
Right Panel: Strengths, weaknesses, scout notes

15. API Endpoints

POST /api/players
POST /api/valuation/generate
GET /api/valuation/compare
GET /api/valuation/report/:playerId

16. Recommended Tech Stack

Frontend: React + Tailwind
Backend: Node.js/NestJS
Database: PostgreSQL
ORM: Prisma
AI Layer: OpenAI
Charts: Recharts

17. Final Product Goal

The MVCE should become the African grassroots football market intelligence layer that predicts market value readiness before the global market notices the player.